UNIVERSITY EXAMINATION BOARD

CALCULUS & DIFFERENTIAL EQUATIONS

Examination Year: 2026 | Maximum Marks: **75** | Total Questions: **3** | Time Allowed: **3 Hours**

GENERAL INSTRUCTIONS:

1. All questions are compulsory unless specified otherwise.
2. Candidates must show all working, intermediate steps, and derivations to earn full marks.
3. Draw diagrams wherever necessary; label all parts clearly.
4. Figures, graphs and diagrams are provided with the respective question.
5. Use of scientific calculator is permitted unless otherwise notified.

Q.No	1	2	3	Total
Marks	25	20	30	75

Question 1 **[25 Marks]**

Consider the scalar field $u(x,y) = x^2 y + x y^2$. Find the directional derivative of u at the point (1,2) in the direction of the vector **v** = (1,1). Show your work including the computation of the gradient ∇u and the unit vector in the direction of **v**.

Question 2 **[20 Marks]**

Solve the first-order ordinary differential equation $(dy / dx) = (x / (x+1)^2)$. Find the general solution and evaluate it at x=0 given the initial condition y(0)=1. Show all steps including separation of variables and integration.

Question 3

[30 Marks]

Solve the second-order linear nonhomogeneous differential equation y'' - 4y' + 3y = e^x. Find the general solution and determine the particular solution satisfying y(0)=1 and y'(0)=0. Show your work including the characteristic equation, complementary solution, and method of undetermined coefficients.